

Branching Out: NLP Plant Recommender

Shubham Banavalikar Ben Robbins Najmeh Rahimi

W266: Natural Language Processing

UC Berkeley School of Information

{shubhambb, ben.robbs, nrahimi}@ischool.berkeley.edu

Abstract

Plant identification is typically approached through images, yet many real-world botanical records and user queries are text-based. We develop an NLP system that predicts plant species from free-text descriptions and compare three model families: TF-IDF retrieval, Sentence-BERT embeddings with cosine similarity, and a domain-adapted BERT model fine-tuned on botanical text. Using 3,000 curated plant entries from the Perennial API for training and independent Wikipedia descriptions for evaluation, we assess each model’s ability to capture botanical terminology and generalize across domains. We imagine a system where a user can describe their perfect garden plant and our system would be able to identify the plant closest to what they are describing. Evaluation is done by generating a user’s potential description using ChatGPT and the wikipedia entry for each plant. We then see if our models are able to pick out the plant based on the generated query as the top result or in the top 5 and 10. Our results improved upon the baseline TF-IDF (0.129 top 10 accuracy) with our domain-adapted BERT model achieving a top 10 accuracy of around 0.190 and the Sentence-BERT embedding model around 0.202. This leaves major room for improvement as the accuracy.

1 Introduction

Text-based plant identification is a challenging and understudied problem. While most consumer and scientific plant identification systems rely on images, many real-world scenarios involve only textual descriptions. Botanists record species traits in field notebooks; gardeners describe the type of plant they want without having a photo; and online plant databases contain rich textual descriptions of morphology, habitat, and care requirements. Enabling a model to infer a plant species directly from such descriptions would provide a practical

tool for classification, recommendation, and knowledge retrieval. This project aims to build a model that takes free-text plant descriptions and predicts the most likely species based on a user’s description of the plant they want. Compared to standard NLP tasks, this domain presents several challenges such as a high variability in text form (expert botanical descriptions vs. casual garden language), domain-specific terminology (e.g., lanceolate, pinnate, glabrous, deciduous), sparse/vague descriptions, and cross-domain robustness as models must generalize beyond the style of their training data. To turn this task into a well-structured experiment, we define the central research question:

Does a domain-fine-tuned BERT encoder outperform SBERT and classical retrieval for predicting plant species from free-text descriptions?

2 Project Overview

2.1 Datasets Used

We initially explored recreating the dataset in (Domazetoski et al., 2025), which relies on species descriptions scraped from the Plants of the World Online (POWO) database. However, this dataset was not publicly released, and reproducing it required extensive web scraping and preprocessing to resolve inconsistencies in formatting, missing fields, and taxonomic structure. Because reconstructing a reliable version of this dataset was beyond the scope of our project timeline, we transitioned to an alternative public source.

Our final training dataset was built using the Perennial Botanical API (per), from which we extracted 3,000 plant entries. While this corpus is relatively small for modern NLP tasks, we found no larger publicly available dataset that provides both reliable species labels and sufficiently detailed textual descriptions. As a result, our experiments reflect the constraints of working with a limited

but high-quality dataset. Each entry includes structured horticultural attributes (e.g., sunlight requirements, watering needs, soil type, growth cycle) as well as a free-text description of the species. The raw dataset contained 1,548 unique common names and 3,000 unique scientific names. Since users of a plant recommendation or identification system are more likely to interact using common names rather than scientific nomenclature, we selected **common names as the prediction labels** for all retrieval-based models.

To evaluate cross-domain generalization, we constructed an independent validation set consisting of Wikipedia-derived plant descriptions. These descriptions differ substantially in writing style and lexical choice from the Perennial dataset, allowing us to test model robustness under more natural and diverse input conditions.

2.2 Background

Domazetoski et al. (Domazetoski et al., 2025) demonstrated that transformer-based models can extract functional plant traits from long botanical descriptions, highlighting the importance of domain-specific vocabulary adaptation for improving model performance. Their findings motivate our hypothesis that fine-tuning BERT on botanical text will outperform general pretrained models for species identification.

Several related works examine recommendation and retrieval using item descriptions. RecoBERT (Malkiel et al., 2020) introduces domain-adapted encoders for catalog item retrieval and shows that fine-tuned BERT models yield substantial improvements in domain-specific semantic understanding.

Across all cited literature, no prior study evaluates species prediction directly from textual plant descriptions as a classification or ranking task. Existing research focuses on trait extraction, embedding analysis, or general recommender systems—not on identifying plant species from free-text descriptions. This project addresses that gap.

2.3 Data Representation and Preprocessing

The Perennial API returns plant information in a nested JSON format containing a mix of free-text descriptions, categorical attributes (e.g., sunlight level, watering frequency, growth cycle), and boolean variables (e.g., drought tolerant, poisonous to pets). To facilitate model training, we flattened the JSON structure into a tabular CSV file in which

each row corresponds to a single plant. The resulting dataset includes both the original description field and a variety of structured metadata capturing morphological, ecological, and horticultural properties.

Although ten plants in the dataset lacked a description field, we retained them because the remaining attributes still contain meaningful biological information. To convert the full set of structured attributes into a form suitable for NLP models, we developed two text-construction pipelines.

In the first approach, we transformed all categorical and boolean fields into human-readable phrases and concatenated them with the original description. This flattened textual representation allowed TF-IDF and transformer-based encoders to consume a unified text field that incorporates both free-text and structured data.

In the second approach, we used a large language model (ChatGPT) to generate a natural-sounding sentence-level summary for each plant. The model received all available structured fields and produced a coherent paragraph describing morphology, growth requirements, and ecological traits. These generated descriptions served as alternative inputs for evaluating whether LLM-based text synthesis improves retrieval performance compared to rule-based text construction.

Together, these two representations enabled us to investigate how different text formats influence downstream species identification models.

3 Models

We evaluate three retrieval-based models for text-only plant species identification: a TF-IDF baseline, a domain-adapted BERT model (RecoBERT), and an advanced Sentence-BERT encoder. Each model is trained on two versions of the Perennial dataset: (1) a rule-based “flattened” text representation constructed from structured attributes, and (2) an LLM-generated natural-language description synthesized from the same attributes. This allows us to measure not only model quality but also the impact of different text-construction approaches on retrieval accuracy. All models are evaluated on an external validation set derived from Wikipedia plant descriptions, and we report Top-1 and Top-10 accuracy to capture both strict and lenient retrieval performance.

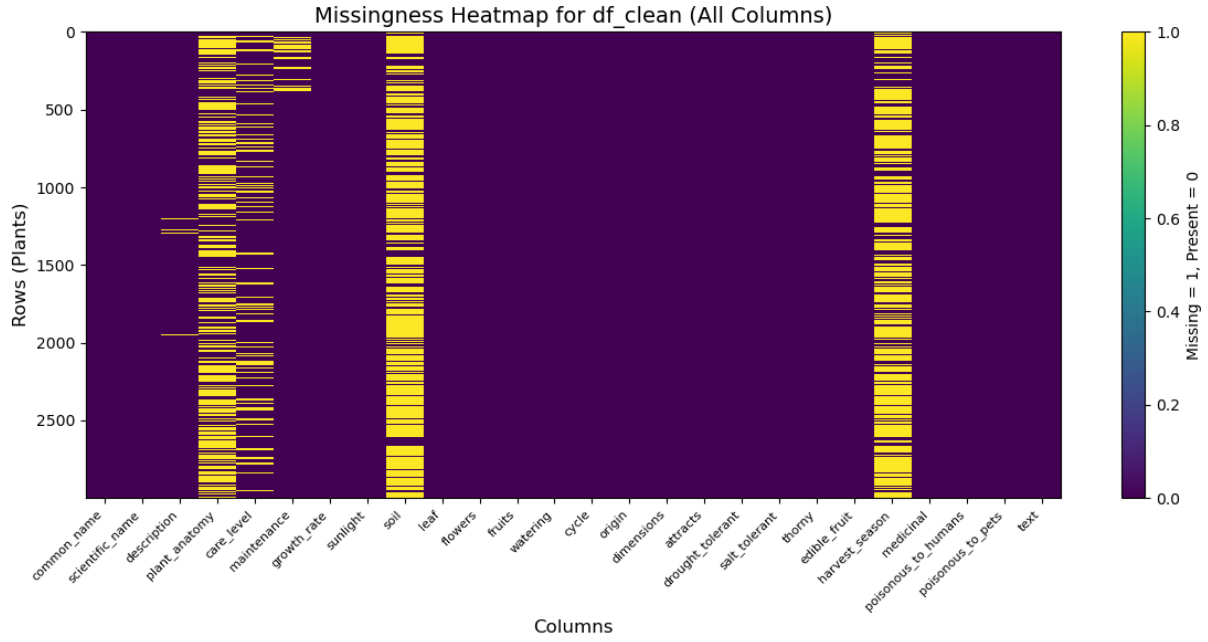


Figure 1: Visualization of missing values across plant attributes in the Perennial dataset.

3.1 Baseline Model

As a classical baseline, we implemented a TF-IDF retrieval model that treats plant identification as a text-matching problem rather than a supervised classifier. Each plant entry in the training set was converted into a single enriched text document combining its natural-language description with structured attributes (sunlight, soil, watering, morphology, boolean traits, etc.). We applied scikit-learn’s `TfidfVectorizer` using unigrams and bigrams to capture both individual botanical terms and short phrases. Test descriptions were vectorized using the same vocabulary, and species predictions were obtained by computing cosine similarity between each query vector and the full set of training vectors. The model returns the most similar species (Top-1) or the top- K most similar candidates, providing a simple and interpretable keyword-based baseline against which the embedding-based models can be compared.

3.2 Reco BERT

RecoBERT has 2 components: title-description model and masked-language model. The title-description model’s objective is to learn whether or not a given plant name and plant description are related. In the paper, they pass tokenized titles and descriptions (with some masked tokens) through `BertModel`, average the title and description embeddings separately, and compute cosine

similarities between the averages. The loss function is defined to optimize for higher cosine similarity between related plant names and descriptions. We also attempted a variation of this where instead of using the embedding averages, we took the `pooler_output` parts of the `BertModel` title and description outputs and computed their cosine similarities. The masked-language model’s objective is to learn embeddings for masked tokens that give high weight to the original token. This allows for incorporating domain-specific nuances. Specifically, it projects `BertModel` embeddings onto the dimension of vocabulary size and uses a loss function that optimizes for a high embedding value for the index corresponding to the original token.

For evaluation, we ranked candidate recommendations by the sum of (1) the cosine similarity between the user query description and the candidate plant name, and (2) the cosine similarity between the user query description and the candidate plant description, where all embeddings used for cosine similarity are produced by the model.

3.3 Advance Sentence BERT

For the implementation of SBert we first used all-MiniLM-L6-v2 from hugging face to encode both the Perennial data and the text generated from the wikipedia descriptions. For each Perennial description embedding a cosine similarity was run against each of the generated descriptions. Then the top k

similar embeddings were selected for each Perennial description where $k = 1, 5$, and 10 . Each of these embeddings were then translated back into their plant names. Accuracy was calculated for each of the k values where the original Perennial name was in the list of predicted values.

In an attempt to improve the accuracy the same model was then finetuned. For this all but the last layer of the model had their parameters frozen. The Perennial dataset text descriptions with the description generated from ChatGPT based on the Perennial dataset were paired with a cosine similarity of 1 . To generate some negative examples the descriptions were shifted 5 places. These were given a cosine similarity of 0 . The model was then finetuned over one epoch using the CoSENTLoss loss function (Reimers and Gurevych, 2023) with a batch size of 16 and a warm up ratio of 0.1 . Code: [SBERT Code](#)

4 Results

For retrieval-style plant identification, the goal is to surface the correct species among a ranked list of candidates, rather than classify every description into a single label. For this reason, to evaluate the various approaches, we decided to use metrics such as **Top-1**, **Top-5**, and **Top-10** accuracy, as they directly measure the quality of ranked recommendations. Traditional metrics like precision and recall assume binary classification and do not reflect how well a model orders candidates or whether the correct plant appears among the top suggestions a user would actually see.

4.1 Accuracy

Model 1	Top-1	Top-5	Top-10
TF-IDF	0.003	0.019	0.042
RecoBERT	0.010	0.050	0.130
Sentence-BERT	0.051	0.133	0.188

Table 1: Retrieval accuracy on the Wikipedia evaluation set using ChatGPT-generated descriptions as the training representation.

Model 2	Top-1	Top-5	Top-10
TF-IDF	0.033	0.085	0.129
RecoBERT	0.010	0.090	0.190
Sentence-BERT	0.050	0.141	0.202

Table 2: Retrieval accuracy on the Wikipedia evaluation set using the rule-based flattened text representation of the Perennial data.

4.2 Comparison of Results

Across both versions of the Perennial dataset discussed at the beginning of the *Models* section, Advanced Sentence-BERT consistently outperforms the TF-IDF baseline and RecoBERT. TF-IDF, while simple and interpretable, shows limited retrieval capacity, although the Top-1 accuracy increases from 0.3% for the generated-descriptions variant to 3.3% for the rule-based variant. RecoBERT performs better than TF-IDF at higher K values, suggesting that its learned alignment between plant names and descriptions provides some semantic signal; however, its Top-1 accuracy remains low, indicating difficulty in selecting the single correct species. Advanced Sentence-BERT shows the strongest performance across all metrics, achieving the highest Top-1, Top-5, and Top-10 accuracies for both Perennial dataset variants. This demonstrates the ability of its sentence encoder to capture richer semantic representations of plant descriptions.

Conclusion

While we were able to improve upon the baseline, our struggle to improve it to a point of getting close to half of the results in the top 10 predicted shows the difficulty of the task. This first and foremost appears to us to be a victim of not enough differentiating information between classes. While we tried to combat a lack of relatively low number of examples per plant species by rewriting the examples with AI, the large number of descriptions combined with the similarity of species made it difficult to differentiate without more information. For example, the SBERT model would predict fig instead of common fig, tall fescue instead of fescue, sweet fennel instead of fennel, and often mix up the many species of apple trees that were in the list. Additionally, having a limited number of descriptions for each plant made it harder for the models to learn unique traits for each of them.

Another thing that may be hindering our results is the prompt phrasings of the generated sets of data. Specifically, generated user versions of plant queries include filler words that distract from the core content to be compared. In our process to build the models we also would occasionally run the models on the raw wikipedia descriptions instead of the generated user versions. These yielded much better results though we chose to exclude them from our results as we thought they were less

authentic to what a human user would query. Given more time and data these models could drastically improve off of the final state of the project.

Authors' Contributions

All team members contributed to dataset exploration, data cleaning, and the construction of both text-based representations (rule-based and LLM-generated). All authors participated in model evaluation, error analysis, and writing the final report. **Najmeh Rahimi** implemented the TF-IDF baseline model. **Ben Robbins** implemented the advanced Sentence-BERT model. **Shubham Banavlikar** implemented the RecoBERT model.

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